Milestone 3 | Detection results

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We tested Aditi's saved YOLOv8n model on the held-out Milestone 2 dataset: 36 images containing 113 labelled faces.

Metric	Baseline	Selected settings
mAP@0.5	96.2%	96.2%
mAP@0.5:0.95	86.2%	86.2%
Precision	89.8%	92.9%
Recall	93.8%	92.0%
Mean IoU of correct matches	0.931	0.932

Confidence cutoffs: baseline 0.25; selected 0.50.

A correct match needs the right identity and box overlap (IoU) of at least 0.50. Mean IoU measures only correct matches, excluding misses and false alarms. mAP uses the full confidence curve, so changing the reporting cutoff alone does not change it.

Results by identity

Identity	Faces	AP@0.5	AP@0.5:0.95	Precision	Recall	Mean IoU
7007	24	93.1%	81.6%	100.0%	62.5%	0.921
2970	23	99.5%	89.1%	100.0%	100.0%	0.937
2336	24	99.5%	89.9%	100.0%	100.0%	0.933
7	23	90.0%	80.3%	82.1%	100.0%	0.932
4428	19	99.0%	90.2%	86.4%	100.0%	0.936

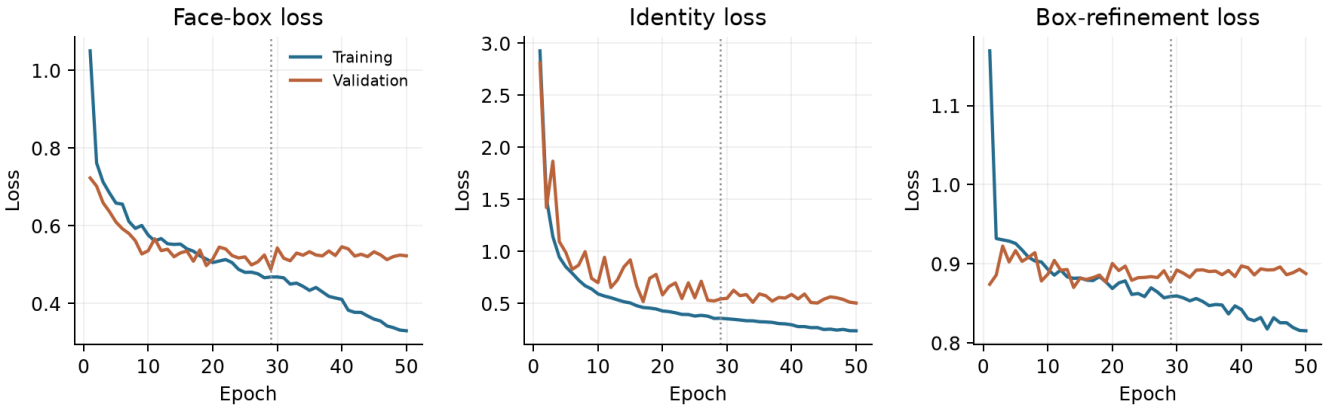
Localization is the strong part. Correct boxes overlap the reference boxes with mean IoU 0.93, and mAP stays at 86.2% even when overlaps up to 0.95 are required. Identity is where the errors are: 8 of the 9 missed faces were found but given the wrong ID, and each of these also counts as one of the 8 false positives. Identity 7007 is affected most (recall 0.62); its faces were labelled 7 (5x) and 4428 (3x). The other identities were all found, but those two IDs collect the confusions, which lowers their precision.

The selected settings produced fewer false alarms, with a small increase in missed faces. The model weights stayed the same; only the prediction settings were compared.

These results apply to synthetic scenes made from a small set of repeated photos. The labels were generated with a face detector. Strong scores here do not guarantee the same performance on real group photos.

Training and validation loss

The model trained for 50 epochs. Epoch 29 had the best validation score and supplied the checkpoint tested here. The test results were not used to choose it.

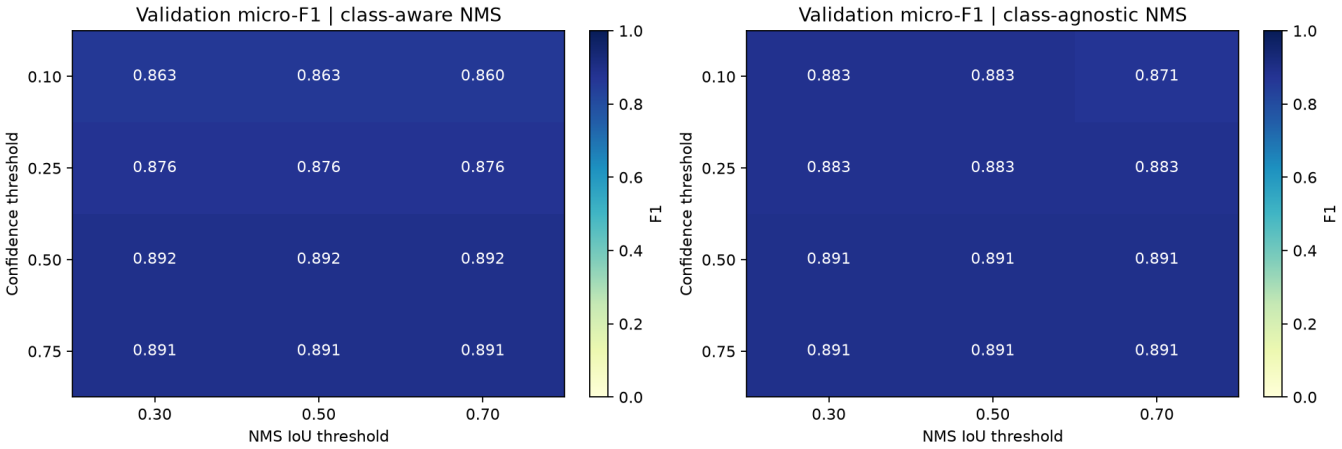


Training loss fell, but validation loss stopped improving as much. The dotted line marks the selected checkpoint.

Hyperparameter comparison

Dario's code compared 24 prediction settings on validation, without retraining. The choice balanced precision and recall using F1. Only the baseline and selected settings were then checked on test.

Setting	Compared	Selected
Confidence cutoff	0.10, 0.25, 0.50, 0.75	0.50
Duplicate-box overlap (NMS IoU)	0.30, 0.50, 0.70	0.70
Duplicate filtering	Within each ID / across IDs	Within each ID

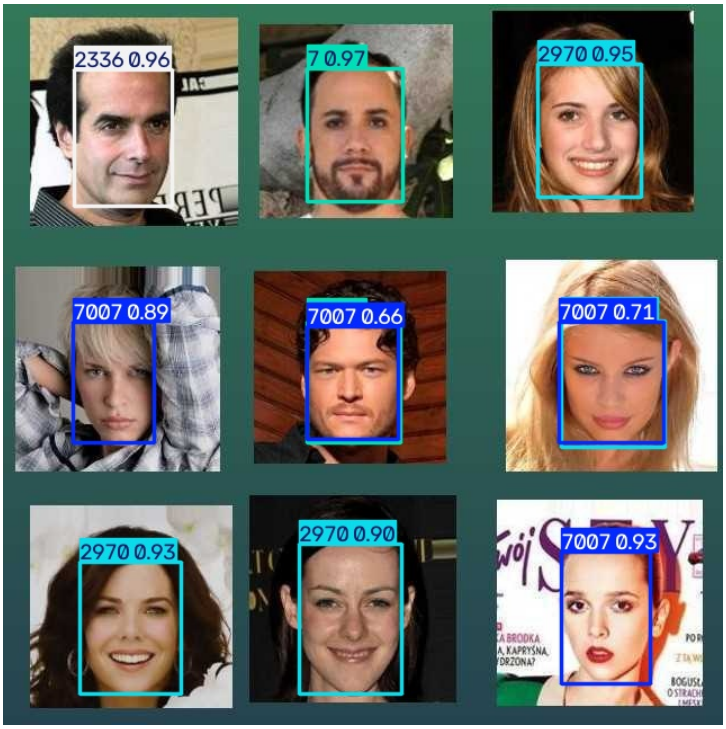


Validation F1 for each confidence cutoff and NMS IoU threshold, with class-aware and class-agnostic NMS.

Raising the confidence cutoff from 0.25 to 0.50 removed 4 false positives on test and lost 2 faces (F1 0.918 to 0.924). On validation, the NMS IoU threshold changed false positives by at most 1 at any cutoff, because faces in our scenes rarely overlap. Class-agnostic NMS lowered validation mAP@0.5 from 97.2% to 87.6%: when the model is torn between two IDs it keeps only one box, and if that box is wrong the face is lost.

TA image check

Using the selected settings, the model placed 11 boxes around all nine faces in the TA image. The middle-centre and middle-right faces each received overlapping predictions with different IDs.



Three tiles are copies of our own training photos, and the model named all three correctly (confidence 0.93-0.96). The other six show people outside our five identities, yet every one received a celebrity ID (8 boxes, confidence 0.58-0.97). Because the ranges overlap, no confidence cutoff separates known from unknown faces. The correct answers also come from photos the model trained on, so they show memorization rather than recognition.

No reference labels were supplied; we identified the three known tiles by comparing each tile with our source photos. The model can output only its trained identities, so Milestone 4 needs a step that can reject unknown faces.

Detailed scores and run settings are saved in project_milestone03/results/.

Metric reference: docs.ultralytics.com/modes/val